Note - Homework 2 - Arduino Pet

Design:

This is a virtual pet running on Arduino UNO, that has a 8x8 LED Matrix for display, potentiometer for temperature control of the pet's virtual environment, photoresistor for sensing when the pet can sleep and two button inputs for feeding and petting. Display shows different responses to its health, energy, hunger stats and player input.

Timing budget:

The system uses timer interrupts every second for sensor reads, some pet updated and counts to 30 seconds for updating pet stats.

The timer ISR and button ISRs are set appropriate flags that are processed in the main loop.

Test results:

Everything seems to be working fine.

Known issues:

Button presses trigger delays for animations that could be improved with flags and processing in the main loop. The project lacks a temperature sensor that would make the use of the potentiometer more appropriate.